UNIVERSITY OF BIRMINGHAM

School of Physics and Astronomy

DEGREE OF B.Sc. & M.Sci. WITH HONOURS

THIRD-YEAR FINAL EXAMINATION

03 00716

LH OBSERVATIONAL COSMOLOGY

SUMMER EXAMINATION 2024

Time Allowed: 1 hour 30 minutes

Answer ***two*** questions. If you answer more than two questions, credit will only be given for the best two answers.

The approximate allocation of marks to each part of a question is shown in brackets **[ ]**.

All symbols have their usual meanings.

Calculators may be used in this examination but must not be used to store text. Calculators with the ability to store text should have their memories deleted prior to the start of the examination.

A formula sheet and two tables of physical constants and units that may be required will be found at the end of this question paper.

ANY CALCULATOR

Answer **two** *questions. If you answer more than two questions, credit will only be given for the best two answers.*

1. (a) Consider an asymptotically de Sitter universe, i.e. a universe that is spatially flat with $\Omega_\Lambda = 1$.

   (i) Derive the acceleration equation for this universe. Is the expansion rate of this universe increasing, decreasing or does it remain constant? **[6]**

   (ii) Find the solution for the scale factor $a(t)$ for this universe. **[6]**

   (iii) Using the fluid equation, show that the equation of state parameter is $w = -1$ for this universe. **[6]**

   (iv) How does such a de Sitter universe compare to our own Universe? **[2]**

   (b) Consider a time-dependent scalar field $\phi(t)$ with a potential $V(\phi) = e^{\alpha\phi(t)}$, where α is a constant, that is the only energy field present in a flat, expanding universe. Derive the Klein-Gordon and Friedmann equations for this cosmological model assuming the slow-roll approximation. **[10]**

2. Consider a flat, matter-only universe with a Hubble parameter $H_0 = 70\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, where decoupling took place at $z_{\mathrm{dec}} \approx 1090$.

   (a) Briefly explain the terms (i) *recombination*, (ii) *surface of last scattering*, and (iii) *decoupling*. **[4]**

   (b) Starting from the Friedmann equation, derive an expression for the time that has passed between two different redshifts, $\Delta t = t(z_1) - t(z_2)$ where $z_1 > z_2$, in this universe. **[10]**

   (c) Calculate the time between decoupling and the present epoch in Gyr. How does this time compare to the age of this universe? **[5]**

   (d) Consider a galaxy observed at a redshift of $z = 2$. How long ago was the light emitted? **[2]**

   (e) Starting from the FLRW metric, derive an expression for the comoving distance, i.e., $r(a)$, travelled by light. **[5]**

   (f) Calculate the comoving and luminosity distance in Mpc to the surface of last scattering in this universe. **[4]**

3. Consider a ΛCDM universe characterised by cosmological parameters that are consistent with current observations of our own Universe.

   (a) Evaluate the value of the dark energy density parameter Ω_Λ when this universe has grown to five times its present day value. **[12]**

   (b) Derive an order of magnitude estimate of how many years after decoupling ($z_{\rm dec} \approx 1100$) the first galaxies formed.
   Hint: Ignoring expansion, the collapse time-scale for non-relativistic matter with an average density $\bar{\rho}$ is

$$\tau_{\rm dyn} \sim (G\bar{\rho})^{-1/2}$$

   [8]

   (c) Matter-radiation equality occurred at a redshift of $z_{\rm eq} \approx 3400$. Assuming that only CMB photons with temperature $T_{\gamma,0} = 2.73$ K and (massless) neutrinos with temperature $T = T_\nu$ contribute to the total radiation density, calculate a blackbody temperature $T_{\nu,0}$ for the neutrinos. Assume that the CMB photon density at the present epoch is 5×10^{-5}. **[10]**

Formula Sheet

FLRW metric in spherical coordinates:

$$ds^2 = -c^2 dt^2 + a^2(t)\left[\frac{1}{1-kr^2/R^2}dr^2 + r^2(d\theta^2 + \sin^2\theta d\phi^2)\right]$$

FLRW metric in alternative coordinates:

$$ds^2 = -c^2dt^2 + a^2(t)R^2\left[d\chi^2 + S_k(\chi)^2(d\theta^2 + \sin^2\theta\, d\phi^2)\right] \text{ with } S_k(\chi) = \begin{cases} \sin\chi & k=+1 \\ \chi & k=0 \\ \sinh\chi & k=-1 \end{cases}$$

Friedmann equation (general case):

$$\left(\frac{\dot{a}(t)}{a(t)}\right)^2 = H^2(t) = \frac{8\pi G}{3}\Big(\rho_r(t) + \rho_m(t)\Big) - \frac{k\cdot c^2}{a^2(t)} + \frac{\Lambda}{3}$$

Friedmann equation (general case) - alternative notation:

$$\frac{H^2(t)}{H_0^2} = \frac{\Omega_{r,0}}{a^4(t)} + \frac{\Omega_{m,0}}{a^3(t)} + \Omega_{\Lambda,0} + \frac{1-\Omega_{\text{tot},0}}{a^2(t)}$$

Acceleration equation (general case):

$$\frac{\ddot{a}(t)}{a(t)} = -\frac{4\pi G}{3}\sum_{i\in\{r,m\}}\left(\rho_i(t) + \frac{3p_i}{c^2}\right) + \frac{\Lambda}{3}$$

Fluid equation (general form) for the i-th density component:

$$\dot{\rho}_i(t) + 3\frac{\dot{a}(t)}{a(t)}(1+w_i)\rho_i(t) = 0$$

Deceleration parameter for N density components:

$$q_0 \equiv -\left(\frac{\ddot{a}}{aH^2}\right)\Bigg|_{t=t_0} = \frac{1}{2}\sum_{i=1}^{N}\Omega_{i,0}(1+3w_i)$$

Planck energy spectrum:

$$E(\nu, T) = \frac{h\nu}{e^{(h\nu)/(k_BT)} - 1}$$

Energy density of a thermal gas:

$$\rho(T) = \frac{8\pi^5 k_B^4}{15h^3c^3}T^4$$

CMB power spectrum:

$$C(\theta) = \sum_{\ell=0}^{\infty} \frac{2\ell+1}{4\pi} C_\ell P_\ell(\cos\theta) \quad \text{and} \quad C_\ell = 2\pi \int_{-1}^{1} C(\theta) P_\ell(\cos\theta)\, d\cos\theta$$

Jeans length:

$$\lambda_J = 2\pi c_s t_{\rm dyn} = 2\pi \left(\frac{2}{3}\right)^{1/2} \frac{c_s}{H} = 2\pi \left(\frac{2}{3}\right)^{1/2} \sqrt{w} \frac{c}{H}$$

Relativistic equation of motion for the growth of small matter perturbations δ:

$$\ddot{\delta}(t) + 2H(t)\dot{\delta}(t) - \frac{3}{2}\Omega_m(t) H^2(t)\delta(t) = 0$$

Saha equation:

$$\frac{1-X}{X^2} = 3.84\eta \left(\frac{k_B T}{m_e c^2}\right)^{3/2} \exp\left(\frac{Q}{k_B T}\right)$$

Neutron-to-proton ratio:

$$\frac{n_n}{n_p} = \exp\left(-\frac{(m_n - m_p)c^2}{k_B T}\right)$$

Klein-Gordon equation:

$$\ddot{\phi}(t) + 3H(t)\dot{\phi}(t) = -\hbar c^3 \frac{dV(\phi)}{d\phi}$$

Astrophysical Constants and Units

Astronomical Unit	AU	1.50×10^{11} m
Parsec	pc	3.1×10^{16} m
Tropical year	y	365.242 mean solar days
Solar luminosity	$L_\odot$	3.84×10^{26} W
Absolute bolometric magnitude of the Sun	$M_{\rm bol}$	$+4^{\rm m}75$
Bolometric correction for the Sun	BC	$-0^{\rm m}08$
Apparent visual magnitude of the Sun	$m_v(\odot)$	$-26^{\rm m}74$
Solar constant	f	1.36×10^3 W m^{-2}
Solar mass	$M_\odot$	1.989×10^{30} kg
Solar radius	$R_\odot$	6.95×10^8 m
Solar effective temperature	$T_{\rm eff}$	5777 K
Wien constant	b	2.898×10^{-3} m K
Hubble constant	H_o	70 km s^{-1} Mpc^{-1}
Distance of the Sun from galactic centre		8.3 kpc
Earth mass	$M_\oplus$	5.972×10^{24} kg
Earth radius	$R_\oplus$	6371 km
Jupiter mass	$M_{\rm Jup}$	1.898×10^{27} kg
Jupiter radius	$R_{\rm Jup}$	71 492 km

Physical Constants and Units

Acceleration due to gravity	g	$9.81\,\mathrm{m\,s^{-2}}$
Gravitational constant	G	$6.674 \times 10^{-11}\,\mathrm{N\,m^2\,kg^{-2}}$
Ice point	T_{ice}	$273.15\,\mathrm{K}$
Avogadro constant	N_A	$6.022 \times 10^{23}\,\mathrm{mol^{-1}}$
		[*N.B.* 1 mole $\equiv 1$ *gram-molecule*]
Gas constant	R	$8.314\,\mathrm{J\,K^{-1}\,mol^{-1}}$
Boltzmann constant	k, k_B	$1.381 \times 10^{-23}\,\mathrm{J\,K^{-1}} \equiv 8.62 \times 10^{-5}\,\mathrm{eV\,K^{-1}}$
Stefan constant	σ	$5.670 \times 10^{-8}\,\mathrm{W\,m^{-2}\,K^{-4}}$
Rydberg constant	R_∞	$1.097 \times 10^{7}\,\mathrm{m^{-1}}$
	$R_\infty hc$	$13.606\,\mathrm{eV}$
Planck constant	h	$6.626 \times 10^{-34}\,\mathrm{J\,s} \equiv 4.136 \times 10^{-15}\,\mathrm{eV\,s}$
$h/2\pi$	$\hbar$	$1.055 \times 10^{-34}\,\mathrm{J\,s} \equiv 6.582 \times 10^{-16}\,\mathrm{eV\,s}$
Speed of light *in vacuo*	c	$2.998 \times 10^{8}\,\mathrm{m\,s^{-1}}$
	$\hbar c$	$197.3\,\mathrm{MeV\,fm}$
Charge of proton	e	$1.602 \times 10^{-19}\,\mathrm{C}$
Mass of electron	m_e	$9.109 \times 10^{-31}\,\mathrm{kg}$
Rest energy of electron		$0.511\,\mathrm{MeV}$
Mass of proton	m_p	$1.673 \times 10^{-27}\,\mathrm{kg}$
Rest energy of proton		$938.3\,\mathrm{MeV}$
One atomic mass unit	u	$1.66 \times 10^{-27}\,\mathrm{kg}$
Atomic mass unit energy equivalent		$931.5\,\mathrm{MeV}$
Electric constant	ϵ_0	$8.854 \times 10^{-12}\,\mathrm{F\,m^{-1}}$
Magnetic constant	μ_0	$4\pi \times 10^{-7}\,\mathrm{H\,m^{-1}}$
Bohr magneton	μ_B	$9.274 \times 10^{-24}\,\mathrm{A\,m^2}$ ($\mathrm{J\,T^{-1}}$)
Nuclear magneton	μ_N	$5.051 \times 10^{-27}\,\mathrm{A\,m^2}$ ($\mathrm{J\,T^{-1}}$)
Fine-structure constant	$\alpha = e^2/4\pi\epsilon_0\hbar c$	7.297×10^{-3}= 1/137.0
Compton wavelength of electron	$\lambda_c = h/m_e c$	$2.426 \times 10^{-12}\,\mathrm{m}$
Bohr radius	a_0	$5.2918 \times 10^{-11}\,\mathrm{m}$
angstrom	Å	$10^{-10}\,\mathrm{m}$
barn	b	$10^{-28}\,\mathrm{m^2}$
torr (mm Hg at $0\,^\circ$C)	torr	$133.32\,\mathrm{Pa}$ ($\mathrm{N\,m^{-2}}$)

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Do not complete the attendance slip, fill in the front of the answer book or turn over the question paper until you are told to do so.

Important Reminders

- Coats/outwear should be placed in the designated area.
- Unauthorised materials (e.g. notes or Tippex) must be placed in the designated area.
- Check that you do not have any unauthorised materials with you (e.g. in your pockets, pencil case).
- Mobile phones and smart watches **must** be switched off and placed in the designated area or under your desk. They must not be left on your person or in your pockets.
- You are not permitted to use a mobile phone as a clock. If you have difficulty seeing a clock, please alert an Invigilator.
- You are not permitted to have writing on your hand, arm or other body part.
- Check that you do not have writing on your hand, arm or other body part – if you do, you must inform an Invigilator immediately
- Alert an Invigilator immediately if you find any unauthorised item upon you during the examination.

Any students found with non-permitted items upon their person during the examination, or who fail to comply with Examination rules may be subject to Student Conduct procedures.